

E A S Y P A N T R Y

Easy Pantry

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Tier 1 — Object Detection

ITAI 1378 — Midterm Blueprint

The Problem

1

Lost track of inventory

Households lose track of pantry inventory, leading to duplicate purchases, forgotten ingredients, and inefficient grocery planning.

2

Why it matters

Families and home cooks care because missing items disrupt meal prep and increase grocery costs.

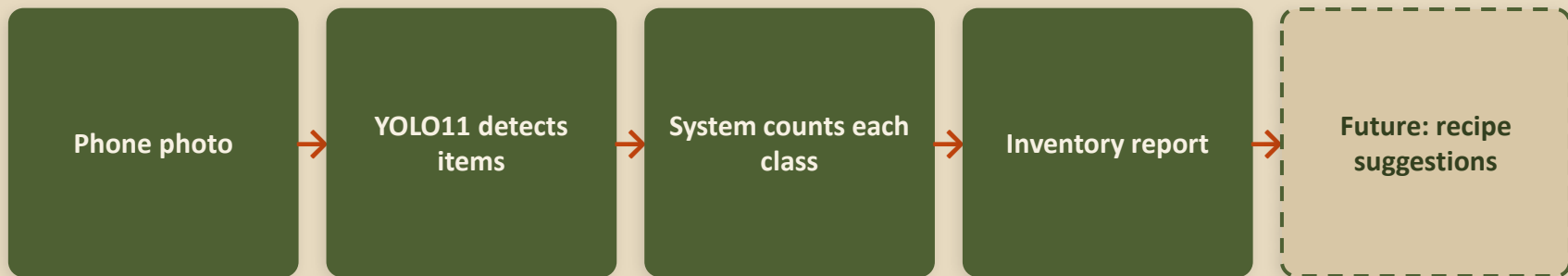
3

The opportunity

A simple automated pantry detection system can save time, reduce waste, and streamline shopping.

Your Solution Workflow

Easy Pantry detects pantry items (pasta boxes, snack bags, soda cans) from a phone photo and generates a quick inventory report.



Step 1 through 4 ship this term; recipe suggestions are a future extension.

Technical Approach

CV TECHNIQUE

Object Detection

MODEL ARCHITECTURE

CNN (YOLO11)

MODEL

YOLO11 (Ultralytics)

USAGE

Pretrained model first → optional transfer learning

FRAMEWORK

PyTorch + Ultralytics

Why: YOLO11 is fast, accurate, and easy to run in Colab — well suited to multi-class pantry detection.

Data Plan

SOURCE	My own pantry and fridge photos, plus optional Roboflow datasets
SIZE	~100 images total
LABELS	pasta_box • snack_bag • soda_can
LABEL STATUS	Personal images labeled in Roboflow; public datasets already labeled

Success Metrics

PRIMARY METRIC

$\text{mAP}_{50} \geq 0.85$

Detection accuracy across all three pantry classes

SECONDARY METRIC

$< 1 \text{ second}$

Inference speed per image, so results feel instant

Milestone Plan — 10-Week Term

Phase	Goal	Milestone	Week
Blueprint	Plan approved	Midterm submitted	Week 5
First working demo	Pretrained YOLO11 runs	Something works	Week 6
Make it yours	Add pantry data + optional training	System works on my data	Weeks 7–8
Improve & measure	Test + metrics	Metrics recorded	Week 9
Package & present	Demo video + README + final slides	Final submitted	Week 10

Risks & Resources

Risk 1 — Not enough labeled data

Plan B: use Roboflow pantry datasets

Risk 2 — Training too slow

Plan B: run pretrained YOLO11 without training

RESOURCES

Compute

Google Colab

Cost

**\$0 — free tier + open
source**

Both risks have a working fallback, so the project ships either way.